EduVision Programming

New Student Assessment Paper

Basic Information

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Current Year: 2025 (Year 7)

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Past experience learning C++ (terms): 0

Past experience learning Python (terms): 3

Part one --- Self-Assessment Trial Questions

These questions are optional. You are encouraged to attempt them. Please submit the source code together with the completed questionnaire.

Problem 1: Numbers Containing Digit 3 or 7

Given two integers m, n, find the number of integers from m to n (inclusive) which contain the digit 3 or 7.

Input Format

One line containing two integers m, n separated by spaces.

Output Format

Output one line containing one integer which is the number of integers from m to n that contain the digit 3 to 7.

Example

Input #1:

```
13  37
```

Output #1:

```
12
```

Explanation

Numbers satisfying the requirements are: 13, 17, 23, 27, 30, 31, 32, 33, 34, 35, 36, 37.

Constraints
$1 \le m \le n \le 10^{18}$
$0 \le n - m \le 10^{7}$

Problem 2: Taxi Fare

Description
A and B are two places d kilometers apart. Bob takes a taxi from A to B at h hours m minutes. Along the way, there are n checkpoints. Checkpoint i is s_i kms from A, the check takes t_i minutes. Taxi fare rules are:

Normal time (7:00–23:00): $3 per km, $2 per waiting minute.
Night time (23:00–7:00): $5 per km, $4 per waiting minute.

The taxi drives at a constant speed of 1 km per minute (ignoring acceleration and deceleration). Write a program to calculate how much Bob pays for the trip.

Input Format

1st line: four integers d, h, m, n, followed by n lines, each containing two integers s, t.

Output Format

One line, containing one integer which is the taxi fare.

Example

Input #1:

```
13  12  0  4
3  3
6  7
10  12
12  5
```

Output #1:

93

Constraints
$0 \leq h < 24, 0 \leq m < 60$
$1 \leq n < d \leq 1000$
$1 \leq t_i \leq 100$
For any $1 \leq i < j \leq n, 1 \leq s_i < s_j < d$

Problem 3: Sum of All Expressions

Problem Description

Given a string consisting of digits '1' to '9', you may insert the operator '+' between digits multiple times (or not at all) to form valid expressions. Calculate the sum of the values of all possible valid expressions.

Input Format

One line, containing a string s composed of digits '1' to '9'.

Output Format

One integer, which is the sum of the values of all valid expressions.

Example

Input #1:

125

Output:

176

Explanation #1:
All possible valid expressions are
125
1 + 25 = 26
12 + 5 = 17
1 + 2 + 5 = 8

Input #2:

```
9999999999
```

Output #2:

```
12656242944
```

Constraints

$1 \le |s| \le 10$

Part two ------- Circle your answer next to each question

Syntax and Standard Library

Input, Output, and Variables:	Not learned	Learned	Proficient
Conditional Branching:	Not learned	Learned	Proficient
Loops:	Not learned	Learned	Proficient
One-Dimensional Arrays:	Not learned	Learned	Proficient
Two-Dimensional Arrays:	Not learned	Learned	Proficient
Character Arrays:	Not learned	Learned	Proficient
Strings:	Not learned	Learned	Proficient
Custom Functions:	Not learned	Learned	Proficient
Struct:	Not learned	Learned	Proficient
Files and Basic I/O:	Not learned	Learned	Proficient
STL Templates:	Not learned	Learned	Proficient

Algorithms and Data Structures (Entry Level)

Enumeration:	Not learned	Learned	Proficient
Simulation:	Not learned	Learned	Proficient
Bubble Sort:	Not learned	Learned	Proficient
Quick Sort:	Not learned	Learned	Proficient
Queues:	Not learned	Learned	Proficient

Stacks:	Not learned	Learned	Proficient
Linked Lists:	Not learned	Learned	Proficient
Trees and Binary Trees:	Not learned	Learned	Proficient
Binary Search:	Not learned	Learned	Proficient
Binary Lifting	Not learned	Learned	Proficient
High-Precision Arithmetic	Not learned	Learned	Proficient
Recursion	Not learned	Learned	Proficient
Greedy Algorithms	Not learned	Learned	Proficient
Depth-First Search (DFS)	Not learned	Learned	Proficient
Breadth-First Search (BFS)	Not learned	Learned	Proficient
Basics of Dynamic Programming (DP)	Not learned	Learned	Proficient
1D Dynamic Programming	Not learned	Learned	Proficient
Knapsack Problems	Not learned	Learned	Proficient
Interval DP	Not learned	Learned	Proficient
Euclidean Algorithm	Not learned	Learned	Proficient
Prime Sieves: Sieve of Eratosthenes & Linear Sieve	Not learned	Learned	Proficient
Sets	Not learned	Learned	Proficient
Addition Principle (Rule of Sum) & Multiplication Principle (Rule of Multiplication)	Not learned	Learned	Proficient
Permutations and Combinations	Not learned	Learned	Proficient
STL	Not learned	Learned	Proficient
Prefix Sum	Not learned	Learned	Proficient
Difference Array	Not learned	Learned	Proficient

Algorithms and Data Structures (Advanced Level)

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Shortest Path	Not learned	Learned	Proficient
Single-Source Second Shortest Path	Not learned	Learned	Proficient
Minimum Spanning Tree (MST)	Not learned	Learned	Proficient
Second Minimum Spanning Tree	Not learned	Learned	Proficient
Topological Sorting	Not learned	Learned	Proficient
Eulerian Path and Eulerian Circuit	Not learned	Learned	Proficient
Bipartite Graph Determination	Not learned	Learned	Proficient
Strongly Connected Components (SCC)	Not learned	Learned	Proficient
Articulation Points & Bridges	Not learned	Learned	Proficient
Union-Find (Disjoint Set Union)	Not learned	Learned	Proficient
Fenwick Tree (Binary Indexed Tree)	Not learned	Learned	Proficient
Segment Tree	Not learned	Learned	Proficient
Splay Tree	Not learned	Learned	Proficient
Treap	Not learned	Learned	Proficient
Center, Diameter and Centroid of a Tree	Not learned	Learned	Proficient
DFS Order and Euler Tour of a Tree	Not learned	Learned	Proficient
Tree Difference, Subtree Sum & Doubling	Not learned	Learned	Proficient
Lowest Common Ancestor (LCA)	Not learned	Learned	Proficient
Discretisation	Not learned	Learned	Proficient
Divide and Conquer	Not learned	Learned	Proficient
Radix Sort	Not learned	Learned	Proficient
Tree DP	Not learned	Learned	Proficient
State Compression DP	Not learned	Learned	Proficient
Common DP Optimisations	Not learned	Learned	Proficient

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Euler's Theorem & Euler's Totient Function	Not learned	Learned	Proficient
Bézout's Identity	Not learned	Learned	Proficient
Modular Multiplicative Inverse	Not learned	Learned	Proficient
Extended Euclidean Algorithm	Not learned	Learned	Proficient
Chinese Remainder Theorem	Not learned	Learned	Proficient
Permutations & Combinations on Multisets	Not learned	Learned	Proficient
Derangements & Circular Permutations	Not learned	Learned	Proficient
Inclusion–Exclusion Principle	Not learned	Learned	Proficient
Catalan Numbers	Not learned	Learned	Proficient
Vectors and Matrices	Not learned	Learned	Proficient
Gaussian Elimination	Not learned	Learned	Proficient
String Matching (KMP)	Not learned	Learned	Proficient